MODULE 15

Ouster OS0 Case Study: Errors and Resolution

This module reads a real datasheet line by line — the **Ouster OS0 Rev8** short-range imaging lidar (datasheet rev. 08/03/2026, model OS0-080) — and turns every specification into numbers a surveyor can use. Datasheet values are quoted as published; where a value had to be read from a chart, or where we make an assumption, we say so. Always check the latest datasheet for your unit.

SIMPLE APPROACH

A datasheet is like the label on a box of crayons: it tells you how many colors, how sharp the tips are, and how long they last. But to know whether you can draw a tiny ant or only a big elephant, you have to do a little math with those numbers. That is what this module does for the OS0 laser scanner.

15.1 The specifications that matter for mapping

Specification	OS0 Rev8 value (datasheet)
Range, 80% target / 10% target	75 m (246 ft) / 35 m (115 ft) (100 klx sunlight, >90% detection, 1024 @ 10 Hz)
Minimum range	0.0 m (0.3 m / 1.0 ft optional, 0.5 m / 1.6 ft default)
Channels (vertical resolution)	32, 64 or 128
Field of view	90° vertical (+45° to −45°), 360° horizontal
Horizontal resolution	512, 1024, 2048 or 4096 columns per rotation
Rotation rate	5–40 Hz (modes: 512 and 1024 @ 10–40 Hz; 2048 @ 10/20 Hz; 4096 @ 5/10 Hz)
Angular sampling accuracy	±0.01° vertical and horizontal
Range resolution	0.1 cm / 0.003 ft (0.8 cm / 0.026 ft in Low Data Rate profile)
Range precision (1σ)	±0.25 cm (0.008 ft) min to ±1.5 cm (0.049 ft) max (beyond 1 m / 3.3 ft, 1024 @ 10 Hz)
Range accuracy	±1.25 cm (0.041 ft) Lambertian; ±2.5 cm (0.082 ft) retroreflective
Returns	up to 2
Laser	865 nm, Class 1; beam 5 mm (0.2 in) at exit; divergence 0.35° (FWHM)
Points per second (max)	10 485 760
Timestamp resolution / latency	< 1 μs / < 10 ms
Time sync	PTP, gPTP, external PPS: < 1 ms error; internal clock: < 20 ppm
Built-in IMU	6-axis (InvenSense IAM-20685HP), 640 / 1280 / 2560 Hz (default)
Data rate, 128 ch @ 1024×10 Hz	43–295 Mbps

15.2 Angular resolution and point spacing

KEY EQUATION — Angular steps of a spinning sensor

$$\Delta\theta_H = \frac{360^\circ}{N_{\text{col}}}, \qquad \Delta\theta_V \approx \frac{\text{vertical FOV}}{N_{\text{ch}} - 1}, \qquad s = \rho\,\Delta\theta\ (\text{rad})$$

What every symbol means

Symbol	Meaning	Units
N_{col}	columns (firings per channel) per rotation: 512–4096	—
N_{ch}	number of channels: 32, 64, 128	—
$\Delta\theta_H, \Delta\theta_V$	horizontal and vertical angular step between samples	deg
s	spacing between neighboring points on a surface facing the sensor	m

(The OS0's actual beam elevation angles can be queried from the sensor's intrinsic calibration and need not be exactly evenly spaced; the formula gives the average.)

Mode	Step	5 m / 16 ft	10 m / 33 ft	20 m / 66 ft	35 m / 115 ft
512 columns	0.703°	6.1 cm / 0.20 ft	12.3 cm / 0.40 ft	24.5 cm / 0.80 ft	43.0 cm / 1.41 ft
1024 columns	0.352°	3.1 cm / 0.10 ft	6.1 cm / 0.20 ft	12.3 cm / 0.40 ft	21.5 cm / 0.71 ft
2048 columns	0.176°	1.5 cm / 0.05 ft	3.1 cm / 0.10 ft	6.1 cm / 0.20 ft	10.7 cm / 0.35 ft
4096 columns	0.088°	0.8 cm / 0.03 ft	1.5 cm / 0.05 ft	3.1 cm / 0.10 ft	5.4 cm / 0.18 ft
Vertical, 128 ch	0.709°	6.2 cm / 0.20 ft	12.4 cm / 0.41 ft	24.7 cm / 0.81 ft	43.3 cm / 1.42 ft
Vertical, 64 ch	1.43°	12.5 cm / 0.41 ft	24.9 cm / 0.82 ft	49.9 cm / 1.64 ft	87.3 cm / 2.86 ft
Vertical, 32 ch	2.90°	25.3 cm / 0.83 ft	50.7 cm / 1.66 ft	101 cm / 3.31 ft	177 cm / 5.81 ft
Beam footprint (FWHM)	0.35°	3.6 cm / 0.12 ft	6.6 cm / 0.22 ft	12.7 cm / 0.42 ft	21.9 cm / 0.72 ft

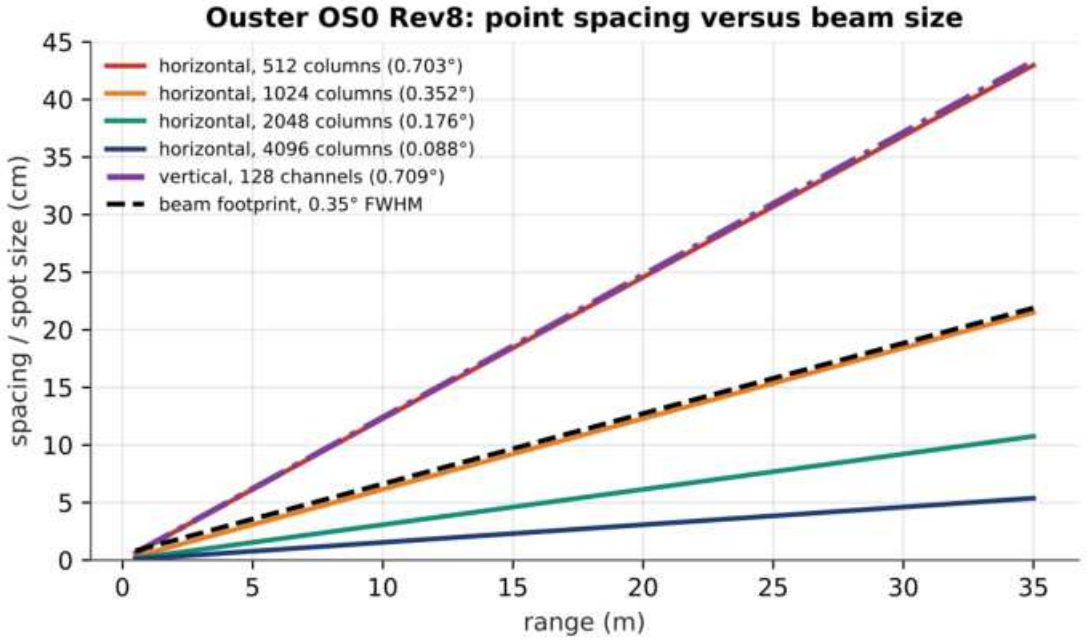


Figure 15.1: Point spacing for each OS0 mode, compared with the size of the laser spot.

WORKED EXAMPLE — spacing on a building face

Given: OS0-128 in 1024 @ 10 Hz mode; building face 20 m away, square to the beam.

Horizontal step

$360/1024 = 0.3516^\circ = 0.006136$ rad

Horizontal spacing

$20 \times 0.006136 = 0.123$ m

Vertical step

$90/127 = 0.7087^\circ = 0.01237$ rad $\Rightarrow 20 \times 0.01237 = 0.247$ m

Result: 12 cm × 25 cm grid in a single sweep. Motion (a walking operator sweeps the scan lines across the wall) fills the vertical gaps over time; that is why mobile clouds are denser than one sweep suggests.

15.3 Beam footprint: the real resolution limit

KEY EQUATION — Footprint from FWHM divergence

$$D(\rho) \approx d_0 + \rho\,\gamma_{\text{FWHM}}, \qquad d_0 = 5 \text{ mm}, \quad \gamma_{\text{FWHM}} = 0.35^\circ = 6.11 \times 10^{-3} \text{ rad}$$

What every symbol means

Symbol	Meaning	Units
γ_{FWHM}	full-width-at-half-maximum divergence: the angle across which the beam's power is at least half its peak	rad
D	spot diameter at range ρ (to half power; some light falls outside it)	m

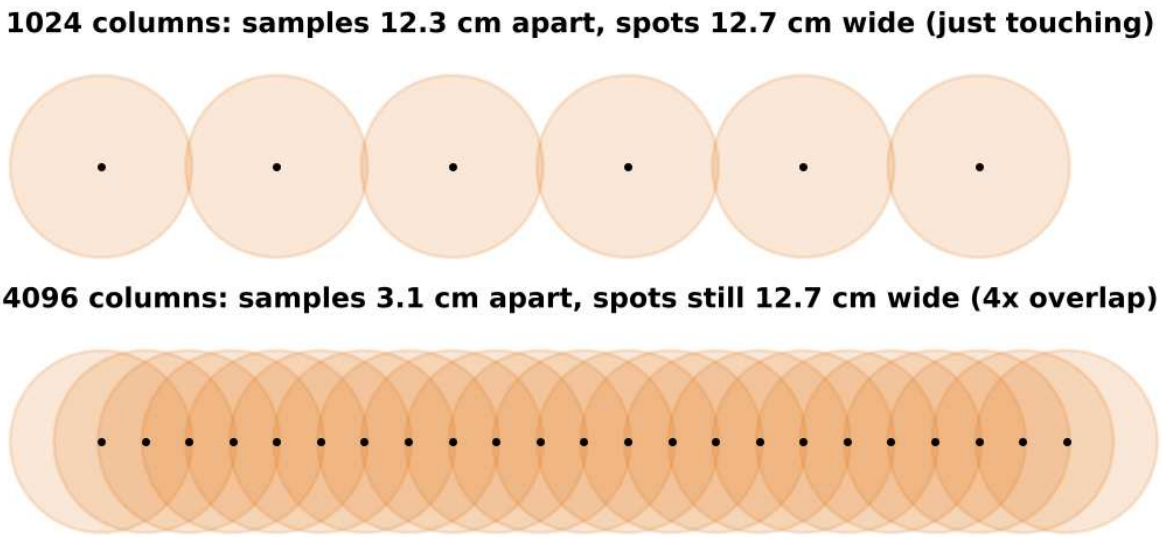


Figure 15.2: At 20 m, the 1024 mode places spots edge to edge; the 4096 mode overlaps each spot with its neighbors.

WORKED EXAMPLE — does 4096 mode make the map sharper?

Spot at 20 m

$0.005 + 20 \times 0.00611 = 0.127$ m

1024-mode spacing

0.123 m — about one spot per spot-width (contiguous coverage)

4096-mode spacing

0.031 m — four samples inside every spot

Result: the 4096 mode gives *more* points but not proportionally *sharper* detail: each sample still averages a 12.7 cm spot. Edges (poles, curb lips) are located only to about ±half a spot, ±6 cm at 20 m. Fine features are best captured *close* to the sensor, where the spot is small.

IMPROVED FROM THE SOURCE MATERIAL

The OS0's 0.35° divergence is about 12 times wider than the 0.3 mrad (0.017°) beam assumed in Module 4 for a survey-grade mapping head. That difference — not the point count — is the main reason short-range robotic lidars and survey scanners produce such different-looking edges. Read the divergence line of any datasheet before reading the point rate.

15.4 Range resolution versus precision versus accuracy

SIMPLE APPROACH

Think of a ruler. **Resolution** is how small the marks are (millimeters). **Precision** is how much your answer wiggles if you measure the same thing ten times. **Accuracy** is whether the ruler itself is the right length. A ruler can have tiny marks and still be stretched!

15.4.1 Resolution (quantization)

KEY EQUATION — Quantization error

$$\sigma_q = \frac{q}{\sqrt{12}}$$

What every symbol means

Symbol	Meaning	Units
q	size of the smallest reported step (the "resolution")	m
σ_q	standard deviation of rounding to that step (a uniform error between $-q/2$ and $+q/2$)	m
$\sqrt{12}$	comes from the variance of a uniform distribution of width q: $q^2/12$	—

WORKED EXAMPLE — does the data profile matter?

Standard profile, $q = 0.1$ cm

$\sigma_q = 0.1/3.464 = 0.029$ cm — negligible against 0.25 cm precision

Low Data Rate profile, $q = 0.8$ cm

$\sigma_q = 0.8/3.464 = 0.23$ cm — *as large as the best precision*

Combined at short range

$\sqrt{0.25^2 + 0.23^2} = 0.34$ cm instead of 0.25 cm

Result: for survey work never use the Low Data Rate profile: it adds a 2.3 mm rounding error to every range, a 36% increase in noise on bright targets.

15.4.2 Precision and accuracy versus range and reflectivity

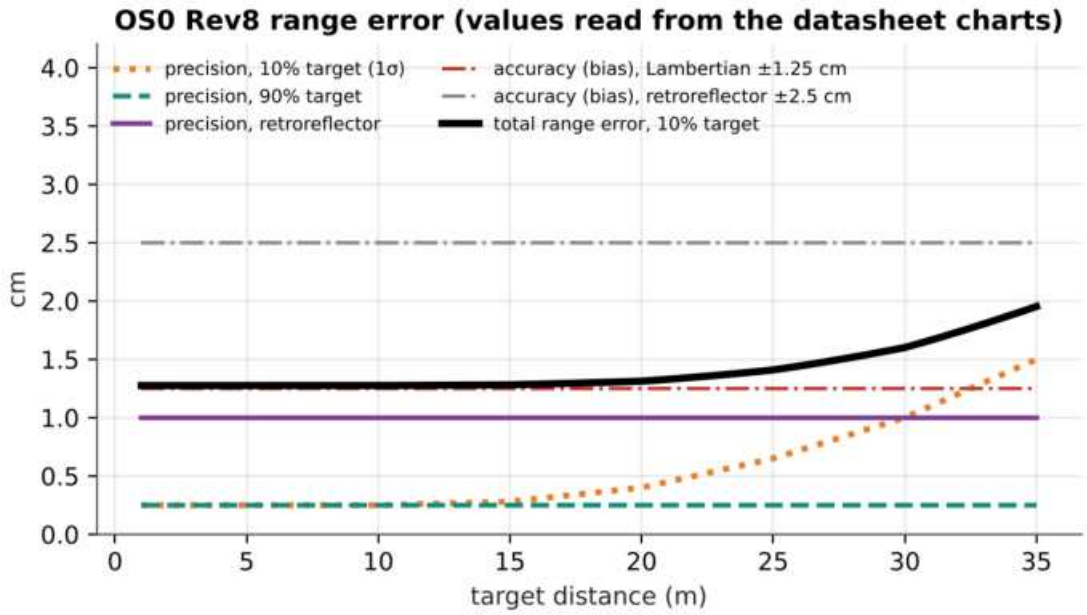


Figure 15.3: Range precision (1σ) and accuracy for the OS0 Rev8. Curves for the 10% target are read from the datasheet chart and are approximate.

Range (m)	5	10	15	20	25	30	35
Range (ft)	16	33	49	66	82	98	115
Precision, 10% target (cm, read from chart)	0.25	0.25	0.28	0.40	0.65	1.00	1.50
Precision, 90% target (cm)	0.25	0.25	0.25	0.25	0.25	0.25	0.25
Precision, retroreflector (cm)	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Accuracy, Lambertian (cm)	1.25	1.25	1.25	1.25	1.25	1.25	1.25
Precision, 10% target (ft)	0.008	0.008	0.009	0.013	0.021	0.033	0.049
Accuracy, Lambertian (ft)	0.041	0.041	0.041	0.041	0.041	0.041	0.041

KEY EQUATION — Total range error from a datasheet

$$\sigma_{\rho,\text{total}} \approx \sqrt{b^2 + \sigma_p^2 + \sigma_q^2}$$

What every symbol means

Symbol	Meaning	Units
b	range accuracy (bias) from the datasheet, treated here as a 1σ systematic error (a conservative assumption; "±" values are not always 1σ)	m
σ_p	range precision at that range and reflectivity	m
σ_q	quantization (usually negligible)	m

WORKED EXAMPLE — total range error on dark asphalt

At 10 m (10% target)

$\sqrt{1.25^2 + 0.25^2 + 0.03^2} = 1.27$ cm

At 20 m

$\sqrt{1.25^2 + 0.40^2} = 1.31$ cm

At 30 m

$\sqrt{1.25^2 + 1.00^2} = 1.60$ cm

At 35 m (maximum range for 10%)

$\sqrt{1.25^2 + 1.50^2} = 1.95$ cm

Result: out to about 25 m the bias dominates and the total barely changes; beyond that, noise on dark surfaces takes over. Because bias is common to nearby points, relative measurements (curb height, cracks) are much better than these totals (Module 14).

SURVEYOR'S FIELD TIP

Retroreflective targets are a trap. The datasheet gives a 2.5 cm accuracy and 1.0 cm precision on retroreflectors — worse than on ordinary surfaces — because the very strong return distorts the pulse timing. Do not use retro-tape targets as checkpoints for absolute height unless the bias has been calibrated; use matte painted targets (Lambertian) and use retro targets for horizontal identification only.

15.5 Angular accuracy

WORKED EXAMPLE — the $\pm 0.01^\circ$ encoder spec

In radians

$0.01^\circ = 1.745 \times 10^{-4}$ rad

At 10 / 20 / 35 m

0.17 / 0.35 / 0.61 cm across the beam (each axis)

Result: smaller than the range error at every range the OS0 can reach — the encoders are not the weak link. (We treat $\pm 0.01^\circ$ as 1σ; if it is a maximum bound, the 1σ is smaller.)

15.6 Maximum range and detection

ENGINEER'S VIEW

The range figures (75 m at 80% reflectance, 35 m at 10%) are for >90% *detection probability* in 100 klx sunlight — they say when points *exist*, not when they are *accurate*. Near the limit the 10% precision is already 1.5 cm and the footprint is 22 cm. Typical asphalt is 5–20% reflective, so on a sunny day expect road returns to thin out beyond roughly 30–35 m. The false-positive rate of 1 in 10 000 means about 131 spurious points per second at 1.31 million points per second: outlier removal (Module 18) is mandatory. With a minimum range setting of 0.0 m, returns from the operator's own body or the vehicle will appear; mask them by azimuth or range.

15.7 Timing, synchronization and skew

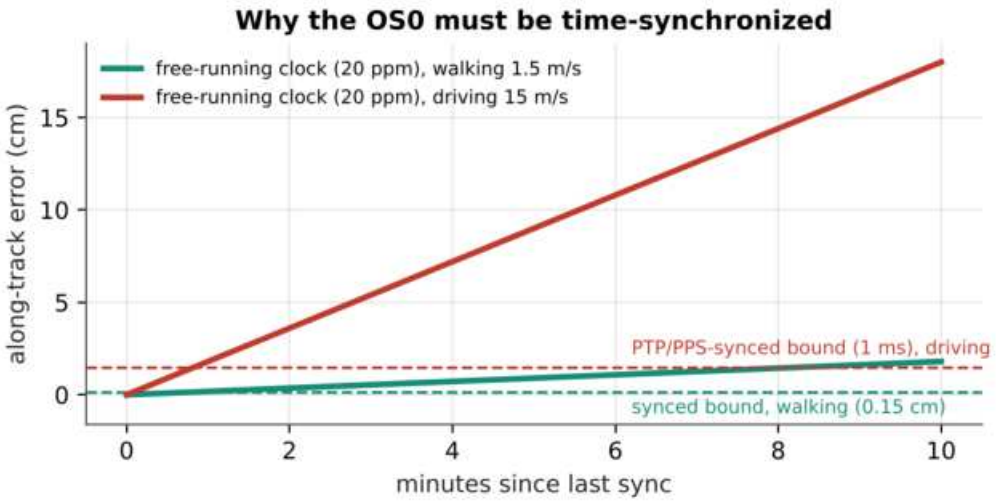


Figure 15.4: Along-track error from timing for a free-running OS0 clock versus a PTP/PPS-synchronized one.

WORKED EXAMPLE — why the OS0 must be synchronized

Given: synced accuracy < 1 ms (PTP or PPS); internal clock drift < 20 ppm.

Synced, walking 1.5 m/s

$1.5 \times 0.001 = 0.15$ cm

Synced, driving 15 m/s

$15 \times 0.001 = 1.5$ cm

Synced, operator turning 90 °/s, point at 20 m

$90^\circ/\text{s} \times 0.001$ s $= 0.09^\circ = 1.57 \times 10^{-3}$ rad; $20 \times 1.57 \times 10^{-3} = 3.1$ cm

Free-running 10 minutes

$600 \times 20 \times 10^{-6} = 0.012$ s; at 15 m/s $\Rightarrow$ 18 cm

Result: the < 1 ms spec is fine for walking, marginal for driving, and poor for fast turns at range. For survey work, synchronize with PPS/PTP to the best your timing source can deliver, and verify with the turn-signature test of Module 8.

WORKED EXAMPLE — scan skew inside one rotation

Given: 1024 @ 10 Hz.

Time per rotation

$1/10 = 0.1$ s; time per column $= 0.1/1024 = 97.7$ μs

Walking at 1.4 m/s

the sensor moves $1.4 \times 0.1 = 0.14$ m between the first and last column

Turning 60 °/s

the head rotates 6° in one sweep; a point at 20 m would smear $20 \times 0.105 = 2.1$ m if not de-skewed

Result: per-column timestamps (< 1 μs resolution) plus IMU motion make de-skewing possible (Module 8). Raising the rotation rate to 20 Hz halves the skew but halves the columns available at each rate setting.

15.8 Built-in IMU and data volume

ENGINEER'S VIEW

The OS0's internal 6-axis MEMS IMU (InvenSense IAM-20685HP) is sampled up to 2 560 Hz ($\Delta t = 0.39$ ms) and time-stamped on the same clock as the lidar — exactly what tightly-coupled SLAM such as LIO-SAM needs for de-skewing and preintegration (Modules 7, 12). Its rigid offset to the lidar is reported through the queryable "IMU pose correction matrix," a factory lever arm and boresight. Its bias and noise are those of an automotive/industrial MEMS part (read its own datasheet for values); expect it to need LiDAR and control aiding continuously, not to coast. Survey backpacks often add an external tactical IMU and GNSS.

Data volume. At 128 channels, 1024 @ 10 Hz: $128 \times 1024 \times 10 = 1\,310\,720$ points/s; maximum data rate 295 Mbps $= 36.9$ MB/s $\approx$ **133 GB per hour**. The quoted maximum 10 485 760 points/s equals $128 \times 2048 \times 20 \times 2$ — consistent with the 2048 @ 20 Hz mode with both returns enabled (our interpretation).

15.9 Putting it together

In a complete backpack system the OS0's own range error is only one term. The full budget — trajectory, attitude, boresight, lever arm, range and encoders — is worked in Module 14 (Section 14.11.1): about **2.5 cm horizontal / 2.0 cm vertical at 5 m, 3.1 / 2.2 cm at 20 m and 4.3 / 2.5 cm at 35 m** (0.08/0.07, 0.10/0.07, 0.14/0.08 ft, 1σ).

15.10 Choosing a mode: a worked decision

WORKED EXAMPLE — interior as-built of a mechanical room

Requirement: 5 cm point spacing on walls up to 15 m; 2 cm (1σ) accuracy; walking survey.

Needed step

$0.05/15 = 0.00333$ rad $= 0.19° \Rightarrow$ 2048 mode (0.176°) or finer

Rotation rate

2048 mode allows 10 or 20 Hz; choose 10 Hz (more points per rotation, still low skew at walking speed)

Footprint check

at 15 m: $0.005 + 15 \times 0.00611 = 0.097$ m — edges will be soft at ± 5 cm; capture pipe and valve details from closer than 5 m (spot 3.6 cm)

Range error

indoor bright walls: $\sqrt{1.25^2 + 0.25^2} = 1.27$ cm; budget with indoor SLAM trajectory (assume 1.0 cm) and 0.03° heading: at 15 m $\sqrt{1.0^2 + 0.79^2 + 1.27^2 + 0.26^2 \cdot 2} = 1.8$ cm ✓

Data

2048 @ 10 Hz $\approx$ 2.6 M points/s, about 2$\times$ the 1024 data volume: plan $\sim$270 GB per hour

Result: 2048 @ 10 Hz, 128 channels, standard (not low-data-rate) profile, PTP sync, scan within 15 m and close up for fine detail. Control: at least three matte targets per room plus checkpoints.

TRY IT YOURSELF

15.1 In 512 mode, what is the horizontal spacing at 8 m?
15.2 What is the OS0 spot diameter at 3 m?
15.3 Compute the total range error for a 10% target at 25 m.
15.4 An unsynchronized OS0 runs for 30 minutes on a vehicle at 10 m/s. Estimate the worst-case along-track error from clock drift alone.

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," of the Reference Book, starting on page 158.

SOURCES FOR THIS MODULE

Ouster, *OS0 Short-Range High-Resolution Imaging Lidar, Rev8 datasheet* (REV 08/03/2026); precision values for 10% targets read from its precision chart.